De La Salle University - Dasmariñas
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS
First Semester, SY 2024-2025

COURSE CODE	**:**	**S-ITPC312Lc**
COURSE TITLE	**:**	**Cisco Networking Semester 1**
COURSE TYPE	**:**	**Lecture**
COURSE CREDIT	**:**	**Two (2) units**
LEARNING MODALITY	**:**	**Flexible Blended Learning Model**
PRE-REQUISITES	**:**	**S-ITCS214/LA**
CO-REQUISITES	**:**	**S-ITPC312L**
PROFESSOR	**:**	**Maria Gloria Del Rosario**
CONSULTATION HOURS	**:**	**Thursday 10:00-12:00, 1:00-3:00 COS100C**
CONTACT INFORMATON:		**madelrosario@dlsud.edu.ph (046)4811900 local 3134**

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Introduction to Networks (ITN) covers the architecture, structure, functions and components of the Internet and other computer networks. Students achieve a basic understanding of how networks operate and how to build simple local area networks (LAN), perform basic configurations for routers and switches, and implement Internet Protocol (IP).

COURSE LEARNING OUTCOME

By the end of this course, students can:

CLO1. configure switches and end devices to provide access to local and remote network resources.
CLO2. explain how physical and data link layer protocols support the operation of Ethernet in a switched network.
CLO3. configure routers to enable end-to-end connectivity between remote devices. CLO4. create IPv4 and IPv6 addressing schemes and verify network connectivity between devices.
CLO5. explain how the upper layers of the OSI model support network applications.
CLO6. use security best practices to configure a small network.
CLO7. Troubleshoot connectivity in a small network.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	**Topic Learning Outcomes**
Midterm Period	
CLO1. Configure switches and end devices to provide access to local and remote network resources.	TLO1. Explain the advances in modern network technologies.
CLO2. Explain how physical and data link layer protocols support the operation of Ethernet in a switched network.	TLO2. Explain how physical layer protocols, services, and network media support communications across data networks.
CLO3. Configure routers to enable end-to-end connectivity between remote devices.	TLO3. Explain how routers use network layer protocols and services to enable end-to-end connectivity.

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Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy	Technology Enabler		Asynchronous Allocated Time	
Module 1 CLO1 TLO1	1 Aug 19-24	Gospel Reading, Reflection and Prayer Presentation on the Fully online learning model LMS: 1. Schoolbook enlistment 2. MS Teams channel for every synch meeting 3. Netacad Course Experience – Enrollment, Interface, Submission, Getting Used to the Cisco Networking Academy LMS Class Participation 1: Key Terms and Concepts Part 1	MS Teams Schoolbook NetAcad		/	2
Module 1 CLO1 TLO1	2 Aug 26 -31	Gospel Reading, Reflection and Prayer Module 1: Networking Today Objective: Explain the advances in modern network technologies Enabling Activity 1: Research IT and Networking Job Opportunities	MS Teams Schoolbook NetAcad		/	2
Module 2 CLO2 TLO2	3 Sep 2-7	Gospel Reading, Reflection and Prayer Module 2: Protocols and Models Objective: Explain how network protocols enable devices to access local and remote network resources Enabling Activity 2: Research Networking Standards	MS Teams Schoolbook NetAcad	/		2
Module 3 CLO2 TLO2	4 Sep 9-14	Gospel Reading, Reflection and Prayer Module 3: Number Systems Objective: Calculate numbers between decimal, binary and hexadecimal systems Enabling Activity 3: Modules Group 1 Exam	MS Teams Schoolbook NetAcad		/	2
Module 3 CLO2 TLO2	5 Sep 16-21	Gospel Reading, Reflection and Prayer Module 3: Number Systems Objective: Calculate numbers between decimal, binary and hexadecimal systems Class Participation 2: Key Terms and Concepts Part 2		/		2

Module 4 CLO2 TLO2	6 Sep 23-28	Gospel Reading, Reflection and Prayer Module 4: Data Link Layer Objective: Explain how media access control in the data link layer supports communication across networks Enabling Activity 4: Modules Group 2 Exam	MS Sch Net			
Module 5 CLO3 TLO3	7 Sep 30 – Oct 5	Gospel Reading, Reflection and Prayer Module 5: Network Layer Objective: Explain how routers use network layer protocols and services to enable endto-end connectivity Enabling Activity 5: Modules Group 3 Exam	MS Teams Schoolbook NetAcad	/		2
Module 4 CLO 4 TLO 2	8 Oct 7-12	Gospel Reading, Reflection and Prayer Module 5: Network Layer Objective: Explain how routers use network layer protocols and services to enable endto-end connectivity Class Participation 3: Key Terms and Concepts Part 3		/		2
Module 1,2,3,4	9 Oct 14-19	Summative Assessment based on Modules 1-4	MS Teams Schoolbook NetAcad		/	2
			TOTAL	8	10	18

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LEARNING PLAN FOR FINALS:

Course Learning Outcomes	**Topic Learning Outcomes**
CLO4. Create IPv4 and IPv6 addressing schemes and verify network connectivity between devices.	TLO4. Implement a VLSM addressing scheme. TLO5. Implement a subnetted IPv6 addressing scheme.
CLO5. Explain how the upper layers of the OSI model support network applications.	TLO6. Compare the operations of transport layer protocols in supporting end-to-end communication. TLO7. Explain the operation of application layer protocols in providing support to enduser applications.
CLO6. Use security best practices to configure a small network.	TLO8. Configure switches and routers with device hardening features to enhance security.
CLO7. Troubleshoot connectivity in a small network.	TLO9. Implement a network design for a small network to include a router, a switch, and end devices.

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Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy	Technolo Enabler	Asynchronous	Synchronous	Alloted Time
Module 6 CLO4 TLO4	10 Oct 21-26	Gospel Reading, Reflection and Prayer Module 6: IPv4 Addressing Objective: Calculate an IPv4 subnetting scheme to efficiently segment a network Subnetting Scenario Subnet an IPv4 Network Class Participation 1: Key Terms and Concepts Part 4			/	2
Module 6 CLO4 TLO4	11 Oct 28-Nov2	Independent Learning Week		/		2
Module 6 CLO4 TLO4	12 Nov 4-9	Gospel Reading, Reflection and Prayer Module 6: IPv6 Addressing Objective: Explain the need for IPv6 Subnetting Scenario Subnet an IPv6 Network Enabling Activity 1: Modules Group 4 Exam	MS Teams Schoolbook NetAcad		/	2
Module 6 CLO4 TLO4	13 Nov 11-16	Gospel Reading, Reflection and Prayer Module 6: IPv4 Addressing Objective: Calculate an IPv4 subnetting scheme to efficiently segment a network Subnetting Scenario Subnet an IPv4 Network Enabling Activity 2: Teacher-made VLSM	MS Teams Schoolbook NetAcad	/		2
Module 7 CLO4 TLO5	14 Nov 18-23	Gospel Reading, Reflection and Prayer Module 7: IPv6 Addressing Objective: Explain the need for IPv6 addressing Class Participation 2: Key Terms and Concepts Part 5	MS Teams Schoolbook NetAcad		/	2
Module 8 CLO5 TLO6 TLO7	15 Nov 25-30	Gospel Reading, Reflection and Prayer Module 8: ICMP Objective: Use various tools to test network connectivity Class Participation 3: Key Terms and Concepts Part 6	MS Teams Schoolbook NetAcad	/		2

Module 9 CLO5 TLO6 TLO7	16 Dec 2-7	Gospel Reading, Reflection and Prayer Module 9: Transport Layer Objective: Compare the operations of transport layer protocols in supporting endto-end communication Enabling Activity 3: Modules Group 5 Exam	MS Tean Schoolbo NetAcad			
Module 10 CLO6 TLO8	17 Dec 9-14	Gospel Reading, Reflection and Prayer Module 10: Network Security Fundamentals Objective: Configure switches and routers with device hardening features to enhance security Enabling Activity 4: Modules Group 6 Exam	MS Teams Schoolbook NetAcad	/		2
Modules 5,6,7,8	18 Dec 16-21	Summative Assessment: Skills-based Activity based on Modules 5-8	MS Teams Schoolbook NetAcad		/	2
			TOTAL	8	10	18

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FINAL PRODUCT/PERFORMANCE:

All students taking this course must accomplish the following requirements listed below in order to gain the certificate of completion for Introduction to Networks.

- Complete all online chapter exams (Chapter 1-10, PT Practice Skills Assessment) with a minimum grade of 80 for all chapter exams.
- Complete the online final exam with a minimum grade of 80.
- Accomplish the teacher-made test and course feedback online.
- Pass the Hands-on Skills Lab.
- Submit the required PT activities completed to 100%.
- Demonstrate mastery on the interactive activity on key terms and concepts.

GRADING SYSTEM Midterm

Enabling Assessments	-	50%
Class Participation	-	20%
Summative Assessments	-	30%
Total		100%

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Enabling Assessments	-	50%
Class Participation	-	20%
Summative Assessments	-	30%
Total		100%

Overall Final Grade = [(Midterm grade) + (Final term grade)] / 2

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the FlexibleBlended Learning Mode. There will be 50% online and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Online classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered synchronously while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Online Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The online meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend online classes, 40% of absence (from online classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.

d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Call number or e-provider	Reference Material
E-Book 2332775	Davies G. 2019. Networking Fundamentals. Packt Publishing Ltd.
E-Book 3038316	VandenBrink R. 2021. Linux for Networking Professionals. Packt Publishing Ltd.
E-Book 2022991	Abhishek Ratan, Eric Chou, Pradeeban Kathiravelu, & Dr. M. O. Faruque Sarker. (2019). Python Network Programming : Conquer All Your Networking Challenges with the Powerful Python Language. Packt Publishing.
E-Book edspgr.11757827	Cisco Technology, I. (San J. C. (2023). Network security from host and network impersonation.
QA 76.73.P98 .Or82 2019	Sarker, Dr. M. 2019. Learning Python networking : a complete guide to build and deploy strong networking capabilities using Python 3.7 and Ansible. Second edition. UK : Packt Publishing, Ltd.
IEEE Xplore	Chadha, D. 2019 Software-Defined Optical Networks, IEEE. Wiley-IEEE Press
IEEE Xplore	Panek,C. 2020. Networking Fundamentals. Wiley. Wiley Data and Cybersecurity.
Website	Login | Skills for All with Cisco. 2024. Netacadcom. [accessed 2024 Aug 5]. https://lms.netacad.com/mod/lti/view.php?id=82254473.

Prepared: **INFORMATION TECHNOLOGY DEPARTMENT**
1st Semester 2024-2025

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